

8. (a) (i) Light of frequency less than $\frac{\phi}{h}$ cannot eject electrons from a surface of work function ϕ , even if the light intensity is increased. Explain this in terms of photons. [3]

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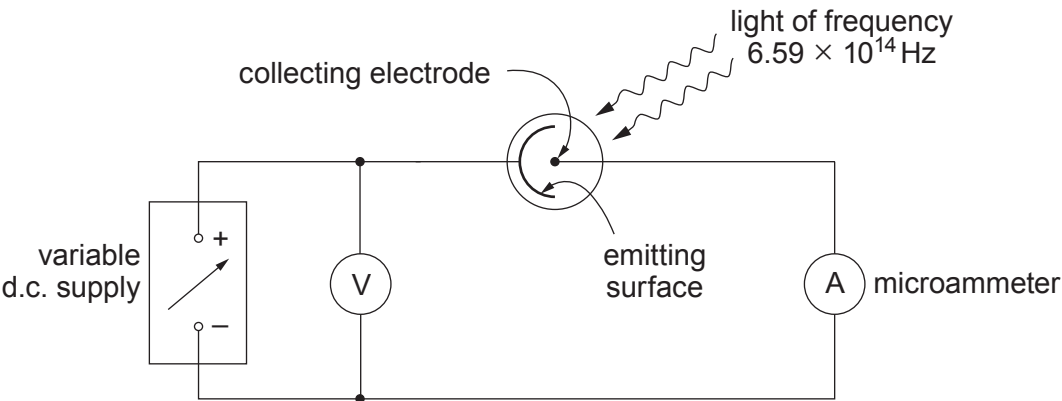
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(ii) The emitting surface in a vacuum photocell is known to be made of one of the metals listed below (with their work functions).

Metal	caesium	potassium	barium	calcium	zinc
Work function / 10^{-19} J	3.12	3.68	4.03	4.59	5.81

The photocell is included in the circuit shown, and illuminated with light of frequency 6.59×10^{14} Hz.



With zero pd applied, the microammeter indicates a current. At some pd between 0 V and 0.35 V the microammeter reading drops to zero.



Determine which metal the emitting surface is made of, giving your reasoning clearly. [4]

Examiner
only

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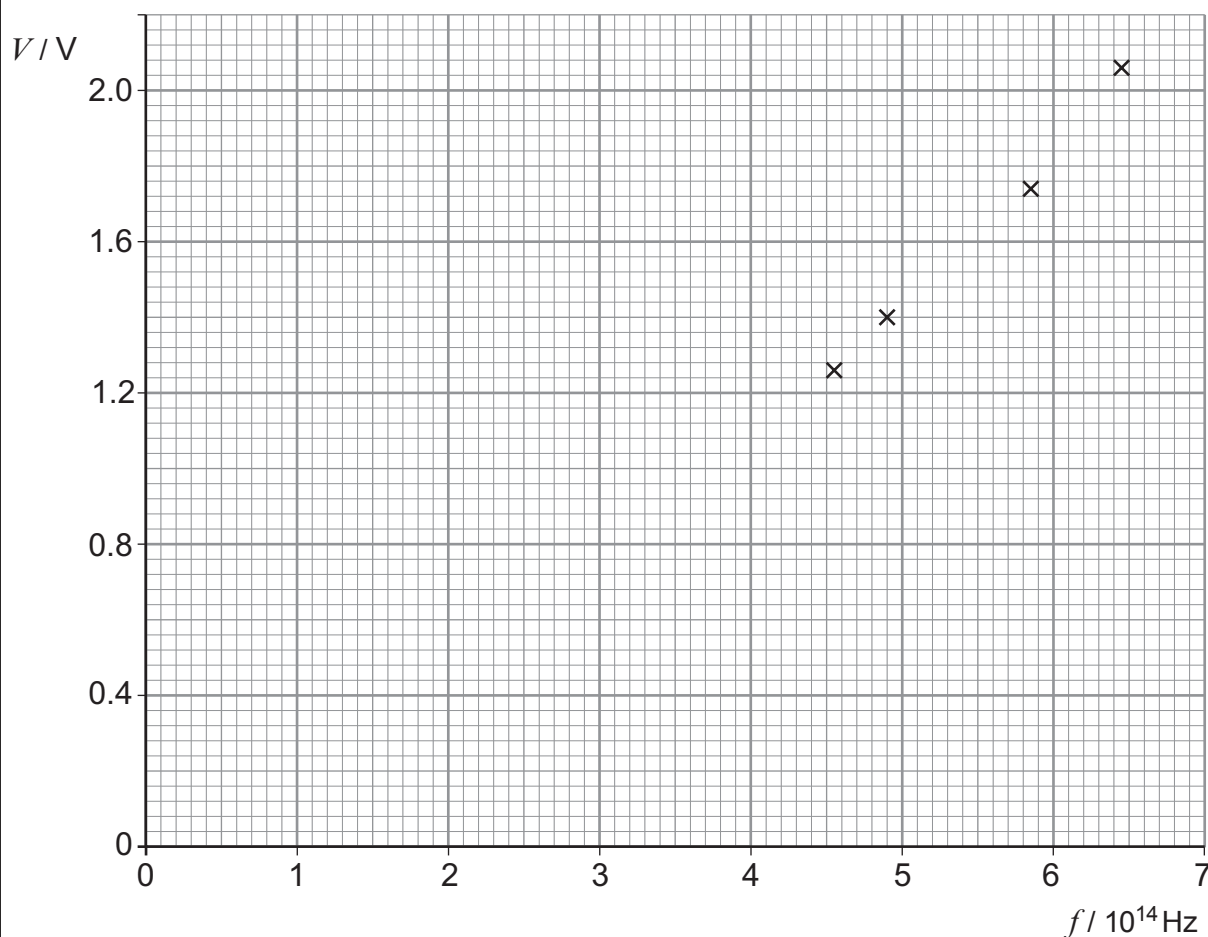
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**TURN OVER FOR THE REST
OF THE QUESTION**



- (b) Rachel varies the pd across a light-emitting diode (LED) and notes the value, V , for which she can just see light from the LED. She also notes the frequency, f , of the light, as supplied by the LED's makers. She does the same for three other LEDs and plots V against f (below).



It has been suggested that V and f are related by the equation:

$$V = \frac{h}{e} f$$

- (i) **On the graph** draw the line of best fit. [1]
- (ii) Discuss the extent to which the graph supports an equation of this form. [2]

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- (iii) Determine the gradient of the graph, and hence a value for h to an appropriate number of significant figures. Assume that the equation predicts $\frac{\Delta V}{\Delta f}$ correctly. [3]
Show your working clearly.

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END OF PAPER

